

## GATE MA

Mathematics

### TREND-BASED MOCK PAPER

— 01 —

|                 |            |
|-----------------|------------|
| Duration        | 3 Hours    |
| Total Questions | 65         |
| Maximum Marks   | 100        |
| Sections        | 6 subjects |

#### Difficulty Distribution

Easy: 27 questions (42%)

Moderate: 32 questions (49%)

Difficult: 6 questions (9%)

# GENERAL INSTRUCTIONS

GATE MA — Trend-Based Mock Paper

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## 1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

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## 2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
  - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
  - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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## 3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
  - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
  - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
  - NAT questions carry no negative marking.
  - MSQ questions carry no negative marking.
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## 4. How to Answer

- For MCQ: Select exactly one option.
  - For MSQ: Select one or more options. Partial credit is NOT awarded.
  - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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## 5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

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## 6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

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## 7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
  - The Question Booklet contains multiple pages. Verify the page count.
  - Immediately report any discrepancy in the Question Booklet to the invigilator.
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**Q1.** 1M [-0.33]

[Algebra]

Rank of 3x3 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

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**Q2.** 1M [-0.33]

[Algebra]

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both +, -, ×
- (D) None

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**Q3.** 1M [no -ve]

[Algebra]

Inverse of AB = ?

Enter your answer here

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**Q4.** 1M [-0.33]

[Algebra]

Identity element property:

- (A)  $a * e = e * a = a$
- (B)  $a * e = e$
- (C)  $e * a = e$
- (D)  $a * e = a$

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**Q5.** 1M [-0.33]

[Calculus]

Limit of  $\sin(x)/x$  as  $x \rightarrow 0$ :

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

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**Q6.** 1M [no -ve]

[Calculus]

Order of  $y'' + 3y' + 2y = 0$ :

Enter your answer here

Q7.

1M [-0.33]

*[Analysis]*

Cauchy sequence:

- (A) Converges always
- (B) Not always
- (C) Never
- (D) Random

Q8.

1M [no -ve]

*[Topology]*

Union of open sets is:

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*Enter your answer here*

Q9.

1M [no -ve]

*[Probability]*

Normal distribution PDF:

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*Enter your answer here*

Q10.

1M [no -ve]

*[Real Analysis]* $f(x)=|x|$  at  $x=0$ :

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*Enter your answer here*

Q11.

1M [-0.33]

*[Algebra]*

Identity element property:

- (A)  $a*e=e*a=a$
- (B)  $a*e=e$
- (C)  $e*a=e$
- (D)  $a*e=a$

Q12.

1M [-0.33]

*[Calculus]*Order of  $y''+3y'+2y=0$ :

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q13.

1M [-0.33]

*[Probability]*

Normal distribution PDF:

- (A)  $(1/\sqrt{2\pi})\exp(-(x-\mu)^2/2\sigma^2)$
- (B)  $\exp(-x)$
- (C)  $1/x$
- (D)  $x^n$

**Q14.**

1M [no -ve]

*[Calculus]*Limit of  $\sin(x)/x$  as  $x \rightarrow 0$ :

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*Enter your answer here***Q15.**

1M [-0.33]

*[Topology]*

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

**Q16.**

1M [-0.33]

*[Algebra]*Rank of  $3 \times 3$  identity:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

**Q17.**

1M [no -ve]

*[Calculus]*Limit of  $\sin(x)/x$  as  $x \rightarrow 0$ :

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*Enter your answer here***Q18.**

1M [-0.33]

*[Topology]*

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

**Q19.**

1M [-0.33]

*[Real Analysis]* $f(x)=|x|$  at  $x=0$ :

- (A) Discontinuous
- (B) Continuous
- (C) Differentiable
- (D) None

**Q20.**

1M [-0.33]

*[Analysis]*

Cauchy sequence:

- (A) Converges always
- (B) Not always
- (C) Never
- (D) Random

Q21.

1M [-0.33]

*[Real Analysis]* $f(x)=|x|$  at  $x=0$ :

- (A) Discontinuous
- (B) Continuous
- (C) Differentiable
- (D) None

Q22.

1M [-0.33]

*[Topology]*

Union of open sets is:

- (A) Open
- (B) Closed
- (C) Neither
- (D) Both

Q23.

1M [-0.33]

*[Analysis]*

Cauchy sequence:

- (A) Converges always
- (B) Not always
- (C) Never
- (D) Random

Q24.

1M [no -ve]

*[Topology]*

Union of open sets is:

---

*Enter your answer here*

Q25.

1M [-0.33]

*[Calculus]*Limit of  $\sin(x)/x$  as  $x \rightarrow 0$ :

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

Q26.

1M [no -ve]

*[Algebra]*

Identity element property:

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*Enter your answer here*

Q27.

1M [-0.33]

*[Algebra]*

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both  $+$ ,  $-$ ,  $\times$
- (D) None

Q28.

1M [-0.33]

*[Calculus]*Limit of  $\sin(x)/x$  as  $x \rightarrow 0$ :

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

Q29.

1M [-0.33]

*[Real Analysis]* $f(x)=|x|$  at  $x=0$ :

- (A) Discontinuous
- (B) Continuous
- (C) Differentiable
- (D) None

Q30.

1M [-0.33]

*[Algebra]*

Identity element property:

- (A)  $a \cdot e = e \cdot a = a$
- (B)  $a \cdot e = e$
- (C)  $e \cdot a = e$
- (D)  $a \cdot e = a$

Q31.

2M [no -ve]

*[Real Analysis]* $f(x)=|x|$  at  $x=0$ :

---

*Enter your answer here*

Q32.

2M [no -ve]

*[Probability]*

Normal distribution PDF:

- (A)  $(1/\sigma\sqrt{2\pi})\exp(-(x-\mu)^2/(2\sigma^2))$
- (B)  $\exp(-x)$
- (C)  $1/x$
- (D)  $x^n$

Q33.

2M [-0.66]

*[Algebra]*

Identity element property:

- (A)  $a \cdot e = e \cdot a = a$
- (B)  $a \cdot e = e$
- (C)  $e \cdot a = e$
- (D)  $a \cdot e = a$

Q34.

2M [-0.66]

*[Algebra]*Inverse of  $AB = ?$ 

- (A)  $B^{-1} A^{-1}$
- (B)  $A^{-1} B^{-1}$
- (C)  $(A+B)^{-1}$
- (D) None

Q35.

2M [no -ve]

*[Probability]*

Normal distribution PDF:

- (A)  $(1/\sqrt{2\pi}) \exp(-(x-\mu)^2/2\sigma^2)$
- (B)  $\exp(-x)$
- (C)  $1/x$
- (D)  $x^n$

Q36.

2M [-0.66]

*[Analysis]*

Cauchy sequence:

- (A) Converges always
- (B) Not always
- (C) Never
- (D) Random

Q37.

2M [no -ve]

*[Calculus]*Order of  $y''+3y'+2y=0$ :

---

*Enter your answer here*

Q38.

2M [-0.66]

*[Algebra]*

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both  $+, -, \times$
- (D) None

Q39.

2M [no -ve]

*[Analysis]*

Cauchy sequence:

- (A) Converges always
- (B) Not always
- (C) Never
- (D) Random

Q40.

2M [no -ve]

*[Topology]*

Union of open sets is:

---

*Enter your answer here*

Q41.

2M [-0.66]

*[Algebra]*

Identity element property:

- (A)  $a*e=e*a=a$
- (B)  $a*e=e$
- (C)  $e*a=e$
- (D)  $a*e=a$

Q42.

2M [no -ve]

*[Algebra]*

Rank of 3x3 identity:

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*Enter your answer here*

Q43.

2M [no -ve]

*[Algebra]*

Identity element property:

- (A)  $a*e=e*a=a$
- (B)  $a*e=e$
- (C)  $e*a=e$
- (D)  $a*e=a$

Q44.

2M [no -ve]

*[Algebra]*

Inverse of AB = ?

---

*Enter your answer here*

Q45.

2M [-0.66]

*[Algebra]*

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both +, -, ×
- (D) None

Q46.

2M [no -ve]

*[Topology]*

Union of open sets is:

---

*Enter your answer here*

Q47.

2M [no -ve]

*[Calculus]*Limit of  $\sin(x)/x$  as  $x \rightarrow 0$ :

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*Enter your answer here*

Q48.

2M [no -ve]

*[Algebra]*

Rank of 3x3 identity:

---

*Enter your answer here*

Q49.

2M [-0.66]

*[Algebra]*

Identity element property:

- (A)  $a * e = e * a = a$
- (B)  $a * e = e$
- (C)  $e * a = e$
- (D)  $a * e = a$

Q50.

2M [-0.66]

*[Algebra]*

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both  $+$ ,  $-$ ,  $\times$
- (D) None

Q51.

2M [no -ve]

*[Topology]*

Union of open sets is:

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*Enter your answer here*

Q52.

2M [no -ve]

*[Analysis]*

Cauchy sequence:

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*Enter your answer here*

Q53.

2M [-0.66]

*[Algebra]*

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both  $+$ ,  $-$ ,  $\times$
- (D) None

Q54.

2M [no -ve]

*[Algebra]*

Rank of 3x3 identity:

---

*Enter your answer here*

Q55.

2M [-0.66]

*[Algebra]*

Identity element property:

- (A)  $a*e=e*a=a$
- (B)  $a*e=e$
- (C)  $e*a=e$
- (D)  $a*e=a$

Q56.

2M [-0.66]

*[Algebra]*

Identity element property:

- (A)  $a*e=e*a=a$
- (B)  $a*e=e$
- (C)  $e*a=e$
- (D)  $a*e=a$

Q57.

2M [-0.66]

*[Algebra]*

Identity element property:

- (A)  $a*e=e*a=a$
- (B)  $a*e=e$
- (C)  $e*a=e$
- (D)  $a*e=a$

Q58.

2M [no -ve]

*[Real Analysis]* $f(x)=|x|$  at  $x=0$ :

---

*Enter your answer here*

Q59.

2M [no -ve]

*[Algebra]*

Identity element property:

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*Enter your answer here*

Q60.

2M [no -ve]

*[Algebra]*

Ring must have:

- (A) 1 only
- (B) 0 and 1
- (C) Both  $+$ ,  $-$ ,  $\times$
- (D) None

**Q61.**

2M [-0.66]

*[Calculus]*Limit of  $\sin(x)/x$  as  $x \rightarrow 0$ :

- (A) 0
- (B) 1
- (C) Infinity
- (D) Undefined

**Q62.**

2M [-0.66]

*[Real Analysis]* $f(x)=|x|$  at  $x=0$ :

- (A) Discontinuous
- (B) Continuous
- (C) Differentiable
- (D) None

**Q63.**

2M [no -ve]

*[Algebra]*Inverse of  $AB = ?$ 

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*Enter your answer here***Q64.**

2M [-0.66]

*[Real Analysis]* $f(x)=|x|$  at  $x=0$ :

- (A) Discontinuous
- (B) Continuous
- (C) Differentiable
- (D) None

**Q65.**

2M [no -ve]

*[Analysis]*

Cauchy sequence:

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*Enter your answer here*